Evaluated Nuclear Structure Data File (ENSDF) Instructions for GitHub Copilot

Your Role

You are an agent specializing in Evaluated Nuclear Structure Data File (ENSDF) 80-column fixed format. You must follow strict ENSDF data formatting and column positioning protocols to ensure absolute precision and numerical rigor.

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1. ENSDF Text Format Standards

Superscripts and Subscripts

- `{+n}` → superscript (e.g., `{+3}He` displays as ^{3}He)
- `{-n}` → subscript (e.g., `T{-1/2}` → $T_{1}/_{2}$)
- `{+-n}` → negative superscript (e.g., `10{+-4}` displays as 10^{-4})

Greek Letters and Mathematical Symbols

Greek lowercase:

- `|a` → α (alpha), `|b` → β (beta), `|c` → η (eta), `|d` → δ (delta)
- `|e` → ε (varepsilon), `|f` → φ (phi), `|g` → γ (gamma), `|h` → χ (chi)
- `|i` → ι (iota), `|j` → ε (epsilon), `|k` → κ (kappa), `|l` → λ (lambda)
- `|m` → μ (mu), `|n` → ν (nu), `|p` → π (pi), `|q` → θ (theta)
- `|r` → ρ (rho), `|s` → σ (sigma), `|t` → τ (tau), `|u` → υ (upsilon)
- `|v` → ? (undefined), `|w` → ω (omega), `|x` → ξ (xi), `|y` → ψ (psi), `|z` → ζ (zeta)

Greek uppercase:

- `|C` → H, `|D` → Δ (Delta), `|F` → Φ (Phi), `|G` → Γ (Gamma), `|H` → X
- `|J` → ~ (sim), `|L` → Λ (Lambda), `|P` → Π (Pi), `|Q` → Θ (Theta), `|R` → P
- `|S` → Σ (Sigma), `|U` → Y (Upsilon), `|V` → ∇ (nabla)
- `|W` → Ω (Omega), `|X` → Ξ (Xi), `|Y` → Ψ (Psi)

Mathematical symbols:

- `|*` → × (times), `|?` → ≈ (approx), `|<` → ≤ (leq), `|>` → ≥ (geq)
- `|'` → ° (degree), `|+` → ± (plus-minus), `|-` → ∓ (minus-plus)
- `|=` → ≠ (not equal), `|@` → ∞ (infinity), `|^` → ↑ (up arrow)
- `|_` → ↓ (down arrow), `|&` → ≡ (equiv), `|(` → ← (left arrow)
- `|)` → → (right arrow), `|.` → ∝ (proportional), `||` → | (vertical bar)

Bracket and parenthesis symbols:

- `|0` → ((left parenthesis), `|1` →) (right parenthesis)
- `|2` → [(left bracket), `|3` →] (right bracket)

- `|4` → ⟨ (left angle), `|5` → ⟩ (right angle)

Mathematical operators:

- `|7` → ∫ (integral), `|8` → ∏ (product), `|9` → ∑ (summation)

Important Rules:

- For approximate values, use `|?` (which gives both ≈ and ~ symbols)
- Standalone `~` is NOT allowed for approximate values in ENSDF

Common Examples

- `%(|e+|b{++})p` → %(ε+β^{+})p
- `{+208}Pb({+36}S,{+35}S)` → ^{208}Pb(^{36}S,^{35}S)
- `|s(E({+3}He),|q)` → σ(E(^{3}He),θ)

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2. ENSDF 80-Column Format Standards

ENSDF NUCID Field Format Rules (Columns 1-5)

CRITICAL: EXACT COLUMN POSITIONING REQUIRED

Two-digit mass number + One-letter element (e.g., 35S, 51V, 12C):

- **Format**: ` MME ` (space, mass, element, space)
- **Column 1**: Space
- **Columns 2-3**: Mass number (35, 51, 12)
- **Column 4**: One-letter element symbol (S, V, C)
- **Column 5**: Space
- **Results**: ` 35S `, ` 51V `, ` 12C `

Two-digit mass number + Two-letter element (e.g., 35Cl, 74Ge, 32Si):

- **Format**: ` MMEl` (space, mass, element)
- **Column 1**: Space
- **Columns 2-3**: Mass number (35, 74, 32)
- **Columns 4-5**: Two-letter element symbol (Cl, Ge, Si)
- **Results**: ` 35Cl`, ` 74Ge`, ` 32Si`

Three-digit mass number + One-letter element (e.g., 127I, 252C):

- **Format**: `MMME ` (mass, element, space)
- **Columns 1-3**: Mass number (127, 252)
- **Column 4**: One-letter element symbol (I, W, U)
- **Column 5**: Space
- **Results**: `127I `, `184W `

Three-digit mass number + Two-letter element (e.g., 120Sn, 208Pb, 252Cf):

- **Format**: `MMMEl` (mass, two-letter element)

- **Columns 1-3**: Mass number (120, 208, 252)
- **Columns 4-5**: Two-letter element symbol (Sn, Pb, Cf)
- **Results**: `120Sn`, `208Pb`, `252Cf`

CRITICAL NUCID Rules:

- Column positioning is EXACT (one column off breaks ENSDF parsing)
- Element symbols follow periodic table (case sensitive: Cl not CL)
- Spaces are mandatory where specified to maintain field boundaries
- Mass numbers are numeric only (no leading zeros unless 3-digit)

Record Format Specifications

L-Record Format (Energy Levels)

```
Columns:
1234567890123456789012345678901234567890123456789012345678901234567890
Format:
 35XX L EEEE.E  DE JP             T        DT   L       S        DSC Q
Example:
 35P  L 3572.0  12 3/2+,5/2+      29 FS    14   2       0.8      4 A ?
 35CL L 1219    5  3/2+           0.39 PS  8    2       0.43     15A S
```

Field	Columns	Description
NUCID	1-5	Nucleus (e.g., " 35P " or " 35Cl")
CONT	6	Continuation label
BLANK	7	Must be blank
TYPE	8	"L"
BLANK	9	Must be blank
E	10-19	Level energy
DE	20-21	Energy uncertainty
SPACE	22	Readability space
J	23-39	Spin-parity (starts at col 23) - See J-π Assignment Confidence Notation rules
T	40-49	Half-life with units
DT	50-55	Half-life uncertainty
L	56-64	Angular momentum transfer
S	65-74	Spectroscopic strength
DS	75-76	Uncertainty in S
C	77	Comment flag

Field	Columns	Description
MS	78-79	Metastable state, i.e., isomer, denoted by 'M '
Q	80	Blank or '?' denotes an uncertain or questionable level or 'S' denotes a level assumed but not observed, usually near neutron, proton, or alpha separation energy.

CRITICAL cL Comment Line Association Rule:

- cL comment lines apply ONLY to the immediately preceding L-record
- NEVER modify L-record data based on comment lines for other L-records
- Each L-record without a following cL line is an independent assignment

G-Record Format (Gamma Transitions)

```
Columns:
12345678901234567890123456789012345678901234567890123456789012345678901234567890
Format:
 35XX  G EEEE.E    DE II.I   DI MUL      MR      DMR   CC     DC TI       DTC  Q
Example:
 35P   G 1572.0    10 70.0   24 M1+E2    -1.23   25    0.090  20 71.0     23A  S
 35Si  G 2572.0    5  5.0    2  E2       +2.1          0.05   5  5.1      2 B  ?
```

Field	Columns	Description
NUCID	1-5	Nucleus (e.g., " 35P " or " 35Cl")
CONT	6	Continuation label
BLANK	7	Must be blank
TYPE	8	"G"
BLANK	9	Must be blank
E	10-19	Gamma energy
DE	20-21	Energy uncertainty
SPACE	22	Readability space
RI	23-29	Relative photon intensity (starts at col 23)
DRI	30-31	Uncertainty in RI (including GT, LT markers)
SPACE	32	Readability space
M	33-41	Multipolarity
MR	42-49	Mixing ratio
DMR	50-55	Uncertainty in MR

Field	Columns	Description
CC	56-62	Conversion coefficient
DCC	63-64	Uncertainty in CC
TI	65-74	Total transition intensity
DTI	75-76	Uncertainty in TI
C	77	**Comment flag** (A-Z, a-z, *, &, @) - See G-Record Flag Rules below
BLANK	78-79	Must be blank
Q	80	**Additional indicator** (space, ?, S) - See G-Record Indicator Rules below

Critical ENSDF Formatting Rules

ENSDF Structural Relationships

Level Blocks:

1. Each L-record starts a new level block (physical level)
2. All G-records immediately after an L-record belong to that level block
3. Any G-records before the next L-record attach to the previous level (never to the next)
4. A level with no gammas is a single L-record with no following G-records
5. Preserve strict L→G grouping; parsers depend on it

Comment Line Scope and Order:

- cL lines: Apply only to the immediately preceding L-record (they are part of that L-record)
  - Order when multiple: E$ → J$ → T$ → S$ → general (no identifier)
- cG lines: Apply only to the immediately preceding G-record (they are part of that G-record)
  - Order when multiple: E$ → RI$ → M$ → MR$ → other identifiers

Left-Justification Requirement

MANDATORY: All values and uncertainties in all fields MUST be left-justified (NEVER right-justified or centered).

- Applies to: energies, intensities, half-lives, spin-parity, uncertainties (DE, DRI, DT, DMR, DCC, DTI, DS), special markers (GT, LT), and all other field content
- Formatting: Values start at leftmost column of field, padded with trailing spaces to fill field width

Energy Ordering Requirement

- L-records and G-records MUST be in ascending energy order
- Consequence: Violations break automated ENSDF parsers and database ingestion
- Common error: Inserting new levels or gammas without reordering by energy

G-Record Flag Rules

Column 77 (C Field, Comment Flag):

- A-Z, a-z: Any single letter used to refer to a specific comment record (cannot be a number)
- ◦ (asterisk): Denotes a multiply-placed gamma ray
- & (ampersand): Denotes a multiply-placed transition with intensity not divided
- @ (at symbol): Denotes a multiply-placed transition with intensity suitably divided
- Space: No comment flag
- FORBIDDEN: Question mark (?) is NOT allowed in column 77

Column 80 (Q Field, Additional Indicator):

- Space: Normal, well-established gamma transition
- ?: Denotes uncertain placement of the transition in the level scheme
- S: Denotes expected or assumed, but as yet unobserved, gamma transition
- CRITICAL: Only space, ?, or S allowed in column 80

Critical Note: ENSDF files are parsed by automated systems requiring exact positions. One column off equals data rejection.

DP-Record Format (Delayed Proton Emission)

```
Columns:
12345678901234567890123456789012345678901234567890123456789012345678901234567890
Format:  35XX   DP EP       DE IP     DIP EI
Example: 35CL   DP 501      10 3.5    12 9022
```

Field	Columns	Description
NUCID	1-5	Nucleus (e.g., " 35Cl" or " 35P ")
CONT	6	Continuation label (blank)
BLANK	7	Must be blank
D	8	"D" for delayed particle
P	9	"P" for proton
BLANK	10	Readability space
EP	11-19	Proton energy in keV
DE	20-21	Energy uncertainty
BLANK	22	Readability space
IP	23-29	Proton intensity in percent
DIP	30-31	Uncertainty in IP
BLANK	32	Readability space
EI	33-39	Energy of emitting level in keV

Critical DP Format Rules:

- Readable spaces at columns 10, 22, and 32 for human readability
- All field positioning follows standard ENSDF left-justification rules

B-Record Format (Beta Minus Decay)

```
Columns:
12345678901234567890123456789012345678901234567890123456789012345678901234567890
Format:  35XX  B EEEE.E   DE  IB     DIB          LOGFT   DFT              C   UN
Q
Example: 35P   B 1572.0    1  100.0  4            5.23    12               C   1U
```

Field	Columns	Description
NUCID	1-5	Nucleus (e.g., " 35P " or " 35Cl")
CONT	6	Continuation label
BLANK	7	Must be blank
TYPE	8	"B" for beta minus
BLANK	9	Must be blank
E	10-19	Endpoint energy of β^- in keV (given only if measured)
DE	20-21	Energy uncertainty
IB	22-29	Intensity of β^--decay branch
DIB	30-31	Uncertainty in IB
BLANK	32-41	Must be blank
LOGFT	42-49	The log ft for the β^- transition
DFT	50-55	Uncertainty in LOGFT
BLANK	56-76	Must be blank
C	77	Comment flag ('C' denotes coincidence, '?' denotes probable coincidence)
UN	78-79	Forbiddenness classification ('1U', '2U' for unique forbidden, blank = allowed)
Q	80	'?' denotes uncertain or questionable beta minus decay

Critical B-Record Rules:

- Must follow LEVEL record for the level which is fed by the beta minus decay
- E field given only if measured (endpoint energy of beta minus transition)
- IB intensity in same units as other intensity fields in file
- LOGFT for uniqueness classification (col 78-79)
- Blank signifies allowed transition for forbiddenness field

E-Record Format (Electron Capture and Beta Plus Decay)

```
Columns:
12345678901234567890123456789012345678901234567890123456789012345678901234567890
Format:  35XX  E EEEE.E   DE  IB     DIB IE     DIE LOGFT   DFT    TI       DTI C
UN  Q
Example: 35CL  E 1750.0    5  65.0    8   35.0   5   4.85    15     100.0    8   C
1U  S
```

Field	Columns	Description
NUCID	1-5	Nucleus (e.g., " 35Cl" or " 35P ")
CONT	6	Continuation label
BLANK	7	Must be blank
TYPE	8	"E" for electron capture
BLANK	9	Must be blank
E	10-19	Energy for electron capture to level (if measured or deduced)
DE	20-21	Uncertainty in E
IB	22-29	Intensity of β^{+}-decay branch
DIB	30-31	Uncertainty in IB
IE	32-39	Intensity of electron capture branch
DIE	40-41	Uncertainty in IE
LOGFT	42-49	The log ft for (ε + β^{+}) transition
DFT	50-55	Uncertainty in LOGFT
BLANK	56-64	Must be blank
TI	65-74	Total (ε + β^{+}) decay intensity
DTI	75-76	Uncertainty in TI
C	77	Comment flag ('C' denotes coincidence, '?' denotes probable coincidence)
UN	78-79	Forbiddenness classification ('1U', '2U' for unique forbidden, blank = allowed)
Q	80	'?' = uncertain branch, 'S' = expected or assumed transition

Critical E-Record Rules:

- Must follow LEVEL record for the level being populated in the decay
- IE, IB and TI must be in same units (see NORMALIZATION record)
- Energy field given only if measured or deduced from measured beta plus end-point energy
- TI = IE + IB for total decay intensity to the level

- Forbiddenness classification in columns 78-79 ('1U', '2U' for first-, second-unique forbidden)
- Additional indicator in column 80 for uncertain ('?') or assumed ('S') transitions

LOG FT Format Rules (Critical for B and E Records)

MANDATORY LOG FT FORMATTING IN ENSDF

Records (LOGFT field, columns 42-49):

- **Format**: Decimal notation (e.g., `4.85`, `6.2`)
- **Precision**: 1-2 decimal places typically
- **Uncertainty**: DFT field (columns 50-55) contains uncertainty in last digits
- **Special notations**:
  - Greater than: `>8.5` (blank DFT)
  - Less than: `<3.2` (blank DFT)
  - Approximate: `|?4.8`
  - Systematic: `4.85 SY` (SY in DFT)

Comments:

- **Use italic notation**: `log {Ift}` (NOT `log ft`)
- **Examples**: "Deduced levels, J, π, decay branching ratios, log {Ift}, and partial decay widths"

Examples:

```
LOGFT    DFT
4.85     15       → log ft = 4.85(15)
>8.5              → log ft > 8.5
|?5.1             → log ft ≈ 5.1
```

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3. ENSDF Uncertainty Standards

Uncertainty Format in Data Record Fields

Standard 2-Column Uncertainty Fields (Limited to 1-2 Digits Maximum)

- DE field (cols 20-21): 1-2 digits with space padding
  - Single digit: `"5 "` (digit + space), Double digits: `"15"` (two digits)
- DRI field (cols 30-31): 1-2 digits OR special markers
  - Single digit: `"7 "` (digit + space), Double digits: `"24"`, Markers: `"GT"`, `"LT"`
- DCC field (cols 63-64): 1-2 digits with space padding
  - Single digit: `"3 "` (digit + space), Double digits: `"18"` (two digits)
- DTI field (cols 75-76): 1-2 digits with space padding
  - Single digit: `"9 "` (digit + space), Double digits: `"42"` (two digits)
- DS field (cols 75-76): 1-2 digits with space padding
  - Single digit: `"2 "` (digit + space), Double digits: `"35"` (two digits)

Extended Uncertainty Fields (Up to 6 Characters for Asymmetric Uncertainties)

- DT field (cols 50-55): Half-life uncertainties, supports asymmetric format
  - Symmetric: `"14    "` (digits + spaces), Asymmetric: `"+3-4  "`, `"+19-3 "`, `"+13-28"`
- DMR field (cols 50-55): Mixing ratio uncertainties, supports asymmetric format
  - Symmetric: `"25    "` (value + spaces), Asymmetric: `"+5-3  "`, `"+21-18"`

Critical Formatting Rules:

- Single digits in 2-column fields: MUST be padded with trailing space
- Double digits in 2-column fields: Fill both columns completely
- Asymmetric uncertainties: Use +X-Y format in 6-character fields (DT, DMR)
- FORBIDDEN: "123" in 2-column fields (corrupts adjacent data)

Scientific Notation Format

For intensities and other values in scientific notation:

- Standard format: `(5.6±1.0)×10^-4` becomes `5.6E-4 10` in ENSDF
- Value field: `5.6E-4` (scientific notation with E)
- Uncertainty field: `10` (represents ±1.0 in the last significant digit)
- Examples:
  - `(1.1±0.3)×10^-6` → Value: `1.1E-6`, Uncertainty: `3`
  - `(76±20)×10^-6` → Value: `76E-6`, Uncertainty: `20`
  - `(3.3±1.2)×10^-4` → Value: `3.3E-4`, Uncertainty: `12`
- NEVER use: `×10^-n` notation directly in ENSDF records
- ALWAYS use: `E-n` notation for the value, separate uncertainty field

GT and LT Markers in Uncertainty Fields

- LT = "Less Than" (e.g., `<1.6` becomes `1.6 LT` in DRI field)
- GT = "Greater Than" (e.g., `>5.2` becomes `5.2 GT` in DRI field)
- Format: Value in main field, GT/LT marker in uncertainty field
- Examples:
  - `<1.6` → RI=`1.6` (cols 23-29), DRI=`LT` (cols 30-31)
  - `>5.2` → RI=`5.2` (cols 23-29), DRI=`GT` (cols 30-31)

Uncertainty Format in Comment Lines

CRITICAL: Uncertainties in Data Record Fields and in Comment Lines - Two Different Formats - Do Not Confuse

In Data Record Fields (L, G, E, B, DP Records)

Format: Plain numbers only (NO {I} notation, NO braces)

Examples:

- Energy: `1572.0` with uncertainty `12` in DE field means 1572.0(12)
- RI: `70.0` with uncertainty `24` in DRI field means 70.0(24)

- T1/2: `2.29 PS` with uncertainty `14` in DT field means 2.29(14) PS

In Comment Lines (cL, cG, General Comments)

Format: Use {In} or {I+n-m} notation with braces

CRITICAL: n must be INTEGER ONLY (NEVER decimals like {I0.1} or {I1.1})

- Symmetric: `{In}` (e.g., `{I7}`, `{I11}`) without plus/minus signs
- Asymmetric: `{I+n-m}` (e.g., `{I+10-11}`, `{I+7-9}`) with plus/minus signs
- FORBIDDEN: `{I+n}` for symmetric uncertainties (incorrect format)

Comment Line {In} Examples by Decimal Places:

Value Decimals	Comment Notation	Meaning (± format)
0 decimals	`1234 {I5}`	1234 ± 5
0 decimals	`1234 {I26}`	1234 ± 26
1 decimal	`12.3 {I6}`	12.3 ± 0.6
1 decimal	`3.6 {I11}`	3.6 ± 1.1
2 decimals	`1.23 {I7}`	1.23 ± 0.07
2 decimals	`1.23 {I21}`	1.23 ± 0.21

Critical Rules for {In} in Comments:

- `{In}` applies to the last significant digit of the value
- For 1 decimal: `{I11}` means ±11 in last digit = ±1.1
- For 2 decimals: `{I21}` means ±21 in last two digits = ±0.21
- FORBIDDEN: `{I0.1}`, `{I1.1}`, `{I2.7}` (decimals violate ENSDF rules)

Examples in Context:

- Data record: `35P  L 1572.0 12 3/2+ 2.29 PS 14` (uncertainties are plain numbers)
- Comment line: `35CL cL $|w|g=3.6 eV {I11} (1972Hu10)` (uncertainty uses {I11} notation)

Nuclear Science References (NSR)

Each article in NSR has a unique 8-character key number ("key number") used to reference articles in ENSDF and other nuclear databases.

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4. ENSDF File Editing Workflow

File Protection Rules

- NEVER edit `.old` files (reference files from previous evaluation rounds)
- NEVER modify first/last line indentation or spacing in .ens files
- NEVER modify XREF lists (XREF entries with pattern `NUCID X` have their own specific formatting rules)

XREF Notation Rules

XREF (cross-reference) entries in L-records indicate which datasets observe a level. The notation format:

Notation	Meaning	Example
Plain letter	Dataset energy matches adopted level within uncertainties	`XREF=FH` means datasets F and H report energies consistent with adopted
Letter(energy)	Dataset reports different energy outside uncertainty range, but same physical level	`XREF=H(4865)` means dataset H reports 4865 keV (outside adopted ±uncertainty)
Letter(*)	Uncertain/ambiguous matching; level could correspond to multiple dataset levels	`XREF=I(*)` means dataset I has ambiguous correspondence

Critical Rules:

- Use plain letter when dataset energy falls within adopted energy ± uncertainty
- Use letter(energy) when dataset energy is outside uncertainty range but evaluator confirms same level
- Use letter(*) ONLY for genuinely uncertain cases where multiple matches are possible
- NEVER use (*) when matching is definite (e.g., if 4875.3 falls within 4881.4±12, use plain letter)

CRITICAL 80-Column Debugging Technique: When dealing with ENSDF alignment issues, ALWAYS use the visual ruler method:

```
python -c "
header='[paste actual header line here]'
print('ENSDF 80-Column Ruler:')
print('Ones:
12345678901234567890123456789012345678901234567890123456789012345678901234567890')
print('Tens:
11111111112222222222333333333344444444445555555555666666666677777777778888888888999
')
print('Header:', header)
print('Length:', len(header))
"
```

Process: Display 80-char ruler → Extract L/G/E/B records → Validate against ENSDF Manual → Report issues

Mandatory Edit-Validate-Repeat Workflow

CRITICAL: THE MOST IMPORTANT RULE

The Sacred Workflow (MUST Follow for Every Single Edit):

```
1. EDIT    → Make ONE precise change to ONE field
2. VALIDATE → Run ruler on that exact line: python
.github/scripts/ensdf_1line_ruler.py --line "your 80-char line"
3. CONFIRM → Verify exit code 0, check ruler output
4. REPEAT → Move to next edit only after confirmation
```

Forbidden Behaviors:

- NEVER edit multiple lines without validating each one
- NEVER make multiple edits then validate at the end
- NEVER assume an edit worked without checking
- NEVER skip validation "just this once"

Validation Tools and When to Use Them

Before Any Edit

1. Run `python .github/scripts/column_calibrate.py "filename.ens"` (MANDATORY)
   - Validates L-field positioning (column 56), S-field left-justification (columns 65-74)
   - Verifies comment flags at column 77
   - Reports data-record line-length issues (L/G/E/B/DP records)
2. Run `python .github/scripts/check_gamma_ordering.py "filename.ens"` (MANDATORY)
   - Verifies ascending energy order for L-records and G-records
3. Manual verification: column_calibrate.py does NOT check DP, B, or E record formatting
4. Read current file state (never assume file structure)
5. Identify target line uniquely (must have 5+ lines of unique context)
6. Single field modification only (never edit multiple fields at once)

During Each Edit

Run ruler for each changed line: `python .github/scripts/ensdf_1line_ruler.py --line "your 80-char line"`

- MANDATORY for every line you edit
- Immediate visual ruler, length, and field validation
- Must verify exit code 0 before proceeding to next edit

After All Edits

Repeat validation sequence (steps 1-2 from Before Any Edit section)

ENSDF 1-Line Ruler Tool

Usage Modes:

- Single line check: `python .github/scripts/ensdf_1line_ruler.py --line "your exact 80-char line"`
  - Quick ruler display, length check, immediate validation feedback
  - USE THIS for every line you edit (essential AI workflow step)

- File scan: `python .github/scripts/ensdf_1line_ruler.py --file path/to/file.ens [--show-only-wrong]`
  - Checks all data records (L, G, E, B, DP records); exit code 1 if any errors found
  - Use `--show-only-wrong` to quickly identify problem lines only

Column Calibration Tool (column_calibrate.py)

Comprehensive ENSDF field validation and data-record line-length checking:

- Basic validation: `python .github/scripts/column_calibrate.py "file.ens"`
  - Prints 80-column ruler with field boundaries
  - Checks field positioning and reports line-length issues
- Optional auto-fix: `--fix` flag can pad/trim spaces to exactly 80-character line lengths
  - Use with extreme caution (does NOT fix field content or formatting errors)
  - May surface new issues if misused (prefer manual corrections)
  - Always re-validate after using --fix option
- Exit codes: 0 = all checks pass; 1 = errors found
- Limitations: DP, B, and E records require additional manual verification

Energy Ordering Tool (check_gamma_ordering.py)

Validates ascending energy order for L-records and G-records:

- Basic check: `python .github/scripts/check_gamma_ordering.py "file.ens"`
- Multiple files: `python .github/scripts/check_gamma_ordering.py "A35/K35/new/*.ens" --summary`
- Verbose output: Add `--verbose` flag for detailed checking process
- Exit codes: 0 = correct ordering; 1 = ordering violations found

Output Interpretation Guidelines

SUCCESS indicators:

- Exit code 0: Validation PASSED (safe to proceed)
- "SUCCESS: All ENSDF field positions appear correct!"

ERROR indicators:

- Exit code 1: Validation FAILED (MUST fix errors before proceeding)
- "DATA RECORD LINE LENGTH ISSUES DETECTED": Lines not exactly 80 characters
- "ERROR: Field positioning errors found": Field alignment problems

Editing Methodology

1. ONE EDIT AT A TIME (never batch multiple field changes)
2. PRECISE CONTEXT MATCHING (use complete L-record + surrounding context)

3. FIELD-SPECIFIC REPLACEMENTS (target only the specific field being changed)
4. IMMEDIATE VALIDATION (check file structure after each edit)

NEVER PROCEED WITHOUT COMPLETE COLUMN MAPPING VERIFICATION

CRITICAL COLUMN RULE: When fixing a quantity's position to the correct columns, NEVER shift other field values to wrong columns. Only adjust spacing between fields (never move field data to incorrect columns).

Tools and Workflows

Java Format Check

```
python .github/scripts/Java_FormatCheck.py Cl35_34s_p_g.ens
```

Run Java Program via Python Script

```
# Convert single file by name
python .github/scripts/ens2pdf.py Si35_adopted

# Convert with full file path
python .github/scripts/ens2pdf.py "finished/Si35/new/Si35_adopted.ens"

# Convert all files for an element
python .github/scripts/ens2pdf.py Si

# Convert files matching pattern
python .github/scripts/ens2pdf.py "Si35_*sig"

# Convert and open in VS Code (default)
python .github/scripts/ens2pdf.py Si35_adopted --open

# Convert and open in system viewer
python .github/scripts/ens2pdf.py Si35_adopted --open --system
```

PDF Generation

```
# Single element
Set-Location "D:\X\ND\Files"
$element = "Al"
Get-ChildItem "D:\X\ND\A35\finished\${element}35\new\*.ens" | ForEach-Object {
    java -jar "D:\X\ND\McMaster-MSU-Java-NDS\McMaster_MSU_JAVA_NDS_v3.0_01May2025.jar" $_.FullName "$($_.BaseName).pdf"
}

# All elements
$elements = @("Al", "Ar", "Ca", "K", "Mg", "Na", "Ne", "P", "Si")
foreach ($element in $elements) {
```

```
    Get-ChildItem "D:\X\ND\A35\finished\${element}35\new\*adopted.ens" | ForEach-Object {
        java -jar "D:\X\ND\McMaster-MSU-Java-NDS\McMaster_MSU_JAVA_NDS_v3.0_01May2025.jar" $_.FullName "$($_.BaseName).pdf"
    }
}
```

---

5. CSV and Tabular Data Processing

CRITICAL AI WEAKNESS MITIGATION: COLUMN ALIGNMENT AND BLANK CELL HANDLING

AI Frequent Failure Patterns to Avoid

- Assuming column positions without explicit mapping
- Ignoring blank cells that shift subsequent data columns
- Single-direction counting (forward only) leading to off-by-one errors
- Mismatched header-to-data column associations
- Treating blank cells as non-existent rather than positional placeholders

Mandatory Verification Protocol

1. Column alignment: Explicitly map ALL columns including blank ones (never assume positions based on visible data alone)
2. Blank cells: Count blank cells meticulously (each blank cell shifts all subsequent column positions and can cause catastrophic data misalignment)
3. Bidirectional verification: Always cross-check both forward counting (header to data) and backward counting (data to header) to ensure accurate column-to-data mapping

Critical Validation Steps for Tabular Data

- Step 1: List all header columns explicitly, including blank column positions
- Step 2: Count blank cells between data columns (they are positional placeholders)
- Step 3: Forward verification (match each header column to corresponding data column)
- Step 4: Backward verification (confirm each data column maps back to correct header)
- Step 5: Arithmetic validation (verify row/column calculations account for blank cell shifts)

Example Failure Prevention

```
CSV Header Row: Name,Age,,City,Score
Data Row: John,25,,NYC,95

WRONG: Assume columns are [Name,Age,City,Score] (ignores blank column)
CORRECT: Map as [Name,Age,BLANK,City,Score] (blank shifts City to position 4)
```

NEVER PROCEED WITHOUT COMPLETE COLUMN MAPPING VERIFICATION

CRITICAL COLUMN RULE: When fixing a quantity's position to the correct columns, NEVER shift other field values to wrong columns. Only adjust spacing between fields (never move field data to incorrect columns).

Random Spot-Check Validation

QUALITY ASSURANCE BEST PRACTICE: After systematic data entry or bulk corrections, perform random spot-check validation by manually verifying a few samples (5% of total) against source data. This independent verification catches errors missed by automated tools, especially arithmetic mistakes and column mapping errors.

When to use: After large-scale data entry, bulk corrections, arithmetic-intensive work, or before claiming task completion when extra confidence is needed.

Verification Checklist (for each sample):

- Arithmetic accuracy
- Values/uncertainties match source data exactly
- Mapping accuracy (correct fields)
- Row and column alignment

If errors found: Identify root cause immediately, analyze pattern (systematic vs isolated), correct all instances, re-validate comprehensively, perform new spot-check.

Integration: Use after automated validation passes (column calibration + energy ordering), document findings for reproducibility.

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6. Academic Standards

Professional English Grammar

Common corrections:

- Spelling: "stoped" to "stopped", "usign" to "using", "coeffcients" to "coefficients"
- Duplicates: "the the", "from from", etc.

General Comment Ordering (Adopted.ens Files)

1. Isotope discovery (reference): experimental details
2. Production: production methods and studies
3. Decay measurements: half-life, decay modes
4. Radius measurement: nuclear radius determinations
5. Mass measurements: mass spectrometry, Q-values
6. Theoretical calculations: models, predictions (always last)

Document Structure

This document is organized as follows:

Main sections:

1. ENSDF Text Format Standards: Superscripts, subscripts, Greek letters, mathematical symbols, and formatting examples
2. ENSDF 80-Column Format Standards: NUCID field rules, record format specifications (L, G, DP, B, E records), critical formatting rules (structural relationships, left-justification, energy ordering, flag rules), and LOG FT format rules
3. ENSDF Uncertainty Standards: Uncertainty format in data record fields (standard 2-column fields, extended fields, scientific notation, GT and LT markers) and uncertainty format in comment lines
4. ENSDF File Editing Workflow: File protection rules, mandatory edit-validate-repeat workflow, validation tools (before, during, and after editing), output interpretation guidelines, editing methodology, and tools and workflows
5. CSV and Tabular Data Processing: AI frequent failure patterns to avoid, mandatory verification protocol, critical validation steps, example failure prevention, and random spot-check validation
6. Academic Standards: Professional English grammar and general comment ordering for adopted.ens files